

25. Elimination Theory and Ideal Theory

J. Ber. d. DMV 33 (1924), pp. 116–120

In what follows I would like to show how elimination theory can be built up in exact parallel with the theory of zeros for polynomials in one indeterminate. This construction rests on a connection between Hentzelt's elimination theory and the methods of field theory, as developed by Steinitz, and ideal theory, as I myself have developed it.¹

I first want to sketch the question in the case of polynomials in one indeterminate. As coefficient domain let a fixed field P be taken once for all, to which the concepts prime function, etc., refer. Here P may be assumed to be abstractly defined in the sense of Steinitz, but in the special case it may naturally coincide with the field of rational numbers or the field of all complex numbers – according to the assumption formerly usual in elimination theory. The theory of zeros for polynomials in one indeterminate now splits into four steps:

1. *The decomposition theorem*, which permits the reduction of the question to one for prime functions – polynomials irreducible in P . For from

$$a(x) = p_1(x)^{e_1} \cdots p_r(x)^{e_r} = q_1(x) \cdots q_r(x)$$

– where the $p(x)$ denote distinct prime functions and the $q(x)$ therefore denote “largest primary functions” – it follows that $a(x)$ vanishes at all and only the zeros of the prime functions $p(x)$.

2. *The construction of the zero field for a prime function $p(x)$* . Such a zero field $\Re = P(\alpha)$ – where α denotes a zero of $p(x)$ – can always, by Steinitz, be constructed as an extension field of P isomorphic to the residue-class field ($\Re$) of $p(x)$. Conversely, every zero field lying in an extension field of P is isomorphic to the residue-class field ($\Re$); zero field and residue-class field are thus abstractly identical.

3. *The decomposition of $p(x)$ into linear factors in the Galois field determined by $p(x)$* . By repeating the construction given under 2. one arrives at the essentially uniquely determined Galois field $P(\alpha_1, \dots, \alpha_m)$, which is precisely sufficient for $p(x)$ to split into linear factors.

4. *The meaning of multiplicity for the original polynomial $a(x)$* . By 1., 2., 3., the question of the zeros, and the corresponding decomposition into linear factors, is settled for an arbitrary polynomial $a(x)$. The multiplicity may be regarded as the degree of the individual primary function $q(x)$ – that is, the number of residue classes modulo $q(x)$ linearly independent with respect to P ; for in the case of the algebraically closed coefficient domain P , where the $p(x)$ become linear, this number is precisely the multiplicity of the zeros.

I now pass to general elimination theory, and therefore take as basis the domain $\mathfrak{S}$ of all polynomials in $x_1, \dots, x_n$ with coefficients from P ; elimination theory can then be defined as the *theory of the zeros of an ideal* from this polynomial domain. By an ideal $\mathfrak{a}$ I understand, as usual, a system of polynomials such that together with f and g also $f + g$, and together with f also Af , belongs to $\mathfrak{a}$, where A is understood to be an arbitrary polynomial from $\mathfrak{S}$. By a zero of $\mathfrak{a}$ I understand every value system $\alpha_1, \dots, \alpha_n$, belonging to an extension field of P , such that $f(\alpha)$ vanishes for every polynomial f belonging to the ideal. If $f_1, \dots, f_t$ denotes an always existing ideal basis of $\mathfrak{a}$ – so that $f = A_1 f_1 + \cdots + A_t f_t$ holds for every f from $\mathfrak{a}$ – then the zeros of $\mathfrak{a}$ are plainly identical with the zeros of $f_1, \dots, f_t$. Since, conversely, every system of finitely many polynomials can be regarded as a basis of the ideal derived from it, the definition of zeros just given for elimination theory is identical with the customary one, but avoids the difficulty lying in the arbitrary singling out of a basis. For polynomials in one indeterminate, where only principal ideals are involved, this difficulty does not arise, since the basis polynomial is uniquely determined up to a unit – a quantity from P as factor – but even there the ideal-theoretic conception will give a more exact insight.

Elimination theory now splits, correspondingly to the case of one indeterminate, into four steps:

I. *The decomposition theorem* – the representation of an ideal by largest primary components – which permits the reduction of the question to one for prime ideals. Here an ideal is called a prime ideal if it divides a product only when it divides at least one factor; it is called a primary ideal if it divides at least one factor or a power of each factor. To every primary ideal $\mathfrak{q}$ there belongs one and only one associated prime ideal $\mathfrak{p}$, which is a divisor of $\mathfrak{q}$ and a power of which is divisible by $\mathfrak{q}$. One has the representation as least common multiple:

$$\mathfrak{a} = [\mathfrak{q}_1, \dots, \mathfrak{q}_r]; \quad \mathfrak{p}_1^{\sigma_1} \cdots \mathfrak{p}_r^{\sigma_r} \equiv 0 \pmod{\mathfrak{a}}.$$

Here the $\mathfrak{q}$ are largest primary components; that is, each $\mathfrak{q}$ is primary, but the least common multiple of any two $\mathfrak{q}$'s is not primary. Moreover, without loss of generality it may be assumed that no $\mathfrak{q}$ divides the

¹Detailed presentation and literature references are in Math. Ann. 90 and in a forthcoming paper.

least common multiple of the remaining ones, so that no $\mathfrak{q}$ can simply be omitted; the $\mathfrak{p}$ are the associated prime ideals of the $\mathfrak{q}$. Now it is not the $\mathfrak{q}$, but – and herein lies the analogue to the case of one indeterminate – their associated prime ideals $\mathfrak{p}$ that are uniquely determined by $\mathfrak{a}$; and since each $\mathfrak{p}$ is a divisor of $\mathfrak{a}$, and conversely a product of powers of the $\mathfrak{p}$ is divisible by $\mathfrak{a}$, the zeros of the ideal therefore coincide with those of its associated prime ideals.

II. *The construction of the zero field for a prime ideal $\mathfrak{p}$.* By definition the residue classes modulo a prime ideal form a ring without zero divisors, containing at least one element distinct from the zero element as soon as $\mathfrak{p}$ is different from the unit ideal; by forming quotients this ring can therefore be extended to the residue-class field $(\mathfrak{R})$ of $\mathfrak{p}$. One can always construct an extension field $\mathfrak{R}$ of P isomorphic to $(\mathfrak{R})$ which becomes the zero field of $\mathfrak{p}$; that is, $\mathfrak{R} = P(\alpha_1, \dots, \alpha_n)$, where $\alpha_1, \dots, \alpha_n$ denotes a zero of $\mathfrak{p}$. Here $\mathfrak{R}$ has transcendence degree k ($0 \leq k < n$) over P , hence contains k elements algebraically independent over P , whereas any $(k + 1)$ elements become algebraically dependent over P . Conversely, every zero field of transcendence degree k lying in an extension field of P is isomorphic to the residue-class field $(\mathfrak{R})$; zero field of transcendence degree k and residue-class field are therefore abstractly identical.

III. *Decomposition of the norm of $\mathfrak{p}$ into linear factors in the Galois field determined by $\mathfrak{p}$.* If the indeterminates $x_1, \dots, x_n$ are regarded as generated from original indeterminates $y_1, \dots, y_n$ by a linear transformation with indeterminates as coefficients, then the zero field $\mathfrak{R}$ of transcendence degree k can be regarded as a finite algebraic extension field of $P(x_{i+1}, \dots, x_n)$ ($k = n - i$). One can therefore pass to the Galois field $\mathfrak{R}$ over $P(x_{i+1}, \dots, x_n)$, in which the prime function $P^{(i)}(x_i, x_{i+1}, \dots, x_n)$ with the zero α_i splits into linear factors. This prime function $P^{(i)}$, when one returns to the original indeterminates y , plays the role of the fundamental equation; for α_i becomes equal to the fundamental form $u_1\beta_1 + \dots + u_n\beta_n$, where β denotes a zero system in the y algebraically dependent on the parameters $x_{i+1}, \dots, x_n$. The decomposition into linear factors thus gives the product over all zeros. But the prime function $P^{(i)}$, or a power of it, may also be regarded as the norm $N(\mathfrak{p})$ of $\mathfrak{p}$, by considering $\mathfrak{p}$ as a module of linear forms in the power products of $x_1, \dots, x_{i-1}$, taking $P(x_{i+1}, \dots, x_n)$ instead of P as coefficient domain, and defining the norm as the product over all elementary divisors – of which only finitely many are distinct from the unit. For a perfect field P as coefficient domain one always obtains $N(\mathfrak{p}) = P^{(i)}$; for an imperfect field a power of $P^{(i)}$ may occur as the norm.

IV. *The norm of the original ideal and the meaning of multiplicity.* If correspondingly one regards the ideal $\mathfrak{a}$ successively as a module of linear forms in the power products of $x_1, \dots, x_{i-1}$ ($i = 1, 2, \dots, n$), then it can be shown that, on taking $P(x_{i+1}, \dots, x_n)$ as basis, this module has in each case only finitely many elementary divisors distinct from the unit; this is a finiteness condition following from the existence of an ideal basis. The product over these finitely many elementary divisors gives the individual norms; the norm $N(\mathfrak{a})$ is defined as the product of all individual norms. The largest primary factors $Q^{(i)}(x_i, x_{i+1}, \dots, x_n)$ of $N(\mathfrak{a})$ correspond one-to-one to the associated prime ideals $\mathfrak{p}$ of $\mathfrak{a}$, in such a way that $Q^{(i)}$ becomes a power of $P^{(i)}$, where $P^{(i)}$ (or, if necessary, a power of $P^{(i)}$) denotes the norm $N(\mathfrak{p})$ of $\mathfrak{p}$. Thus there arises a decomposition

$$N(\mathfrak{a}) = N(\mathfrak{p}_1)^{e_1} \dots N(\mathfrak{p}_r)^{e_r} = Q_1 \dots Q_r,$$

– where if necessary $N(\mathfrak{p})$ is to be replaced by the highest elementary divisor $E(\mathfrak{p}) = P^{(i)}$ – and thus, by I, II, III, the decomposition of the norm into linear factors has been effected for all zeros of $\mathfrak{a}$. The multiplicity of the zeros corresponding to the individual prime ideal $\mathfrak{p}$ of $\mathfrak{a}$ can be regarded as the degree of $Q^{(i)}$, which can be interpreted as the number of residue classes linearly independent over $P(x_{i+1}, \dots, x_n)$. More precisely, these are the residue classes of certain isolated components of the decomposition uniquely determined by $\mathfrak{p}$ and by the prime ideals of higher dimension, namely the residue classes of the ground ideal of the $(i - 1)$ -st stage modulo the component, determined by $\mathfrak{p}$, of the ground ideal of the i -th stage; the latter component has $\mathfrak{p}$ and all prime ideals of higher dimension as associated prime ideals, while the former has only the prime ideals of higher dimension.

A complete *insertion of the case of one indeterminate* is obtained when one passes here too to the principal ideal. The decomposition given under 1. then splits – according to whether $a(x)$ is regarded as basis polynomial or norm – into two separate decompositions: into the representation of the principal ideal as the least common multiple of largest primary components according to I, which here amounts to a representation as a product of powers of prime ideals; or into the decomposition of the norm as a product of powers of the norms of the individual prime ideals according to IV. In the construction of the zero field according to 2., what is in question in reality is the zero field of the prime ideal defined by $p(x)$, while in 3. the norm is decomposed into linear factors. The definition of multiplicity according to 4. is subordinated to that given in IV, since the ground ideal of the zeroth stage is equal to the unit ideal, while for one indeterminate the

ground ideal of the first stage already equals the ideal itself.

The *insertion into the usual elimination theory* is given by the fact that the norm of the ideal has the meaning of a resultant. In this interpretation, the old problem of elimination theory thus becomes a part of the ideal theory of the polynomial domain, namely that part which, beside the decomposition theorems for ideals, also treats the decomposition of their norms and the connection between the two decompositions.

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26. Abstract Construction of Ideal Theory in the Algebraic Number Field

J. Ber. d. DMV 33 (1924), p. 102

4. E. Noether, Göttingen: Abstract construction of ideal theory in the algebraic number field.

The abstract properties were stated that are completely equivalent to ideal theory in the algebraic number field, and that therefore characterize all rings which possess such an ideal theory. They are the rings with identity element and without zero divisors in which the double chain condition holds – every divisor chain of ideals breaks off in the finite, and likewise every multiple chain modulo every ideal different from the zero ideal – and which are integrally closed with respect to their quotient field. If the latter assumption is dropped, then theorems result concerning ideal theory in a finite order, some of which are already found without proof in Dedekind (3rd edition).

A detailed presentation will appear in the *Mathematische Annalen*.

27. Hilbert Numbers in Ideal Theory

J. Ber. d. DMV 34 (1925), p. 101

E. Noether: Hilbert numbers in ideal theory. By Hilbert numbers one generally understands numbers that refer to the enumeration of residue classes of ideals. Hilbert himself, in his “characteristic function,” gave a function which, for the ideals of the polynomial domain, gives from a certain bound onward the exact number of linearly independent residue classes. For the lower degree numbers Macaulay supplemented this function by a generating function, which Ostrowski was able to evaluate explicitly. But Macaulay also gave numbers of another kind, which have proved to be the essential ones for general ideal theory, since they can be formulated abstractly. In accordance with the general decomposition theorems, it is enough to give the definition for primary ideals $\mathfrak{q}$. What is involved is the number of residue classes, linearly independent with respect to the residue-class field of the associated prime ideal $\mathfrak{p}$. These numbers split into partial systems, each given by the number of irreducible components of $\mathfrak{q}, \mathfrak{q}/\mathfrak{p}, \mathfrak{q}/\mathfrak{p}^2, \dots$. The number of the i -th partial system is also characterized by the number of terms of a composition series which runs from $\mathfrak{q}/\mathfrak{p}^i$ to $\mathfrak{q}/\mathfrak{p}^{i-1}$. The total composition series and its Hilbert numbers form the equivalent of the representation of primary ideals as powers of prime ideals, which is no longer valid in the general case – for instance in finite orders of an algebraic number field.

28. Group Characters and Ideal Theory

J. Ber. d. DMV 34 (1925), p. 144

3. E. Noether, Göttingen: Group characters and ideal theory.

The Frobenius theory of group characters – that is, of representations of finite groups – is conceived as the ideal theory of a completely reducible ring, the group ring. Therefore the isomorphism theorems for the additive decomposition of a completely reducible ring are developed in general; the representation as a direct sum of directly indecomposable, simple ideals is unique up to isomorphism, and the indecomposable one-sided ideals belonging to the same two-sided ideal are mutually isomorphic. If therefore isomorphic ideals are combined into one class, then there are as many distinct indecomposable ideal classes as there are two-sided indecomposable ideals.

In the specialization to completely reducible systems of hypercomplex quantities – in particular to the group ring – indecomposable ideal classes and irreducible representation classes correspond mutually and uniquely. Furthermore, two-sided indecomposable ideals and characters correspond; and the number of

additively indecomposable components of a two-sided ideal is equal to the rank of the indecomposable ideals. Thus the Frobenius theory has been classified within this framework.

A detailed presentation is to appear in the Math. Ann.

29. The Finiteness Theorem for the Invariants of Finite Linear Groups of Characteristic p

Nachr. v. d. Ges. d. Wiss. zu Göttingen 1926, pp. 28–35

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I show in what follows that the well-known finiteness theorem for the invariants of finite groups of linear substitutions² remains valid when, as coefficient domain, instead of an ordinary number field one takes an arbitrary field – hence in particular one of characteristic p .³ However, the known proofs of finiteness fail as soon as the order of the group is divisible by p ; deeper arithmetical facts must be brought in, namely the following criterion, previously known only for characteristic zero:

Finiteness criterion⁴: An integral domain $\mathfrak{S}$ of polynomials – more generally of rational functions – with coefficients from a field is finite if and only if $\mathfrak{S}$ contains a finite subring $\mathfrak{R}$ such that $\mathfrak{S}$ depends algebraically and integrally on $\mathfrak{R}$. Each integral basis of $\mathfrak{R}$ is then completed, by a module basis of $\mathfrak{S}$ with respect to a certain subring of $\mathfrak{R}$, to an integral basis of $\mathfrak{S}$.

This criterion – which is of independent interest in itself – rests on the existence of the rational basis and on the fact that the divisor-chain theorem is preserved in passing to the integral quantities of a finite extension. This second theorem was proved in the preceding note of Artin and van der Waerden for arbitrary characteristic, only under a certain finiteness assumption on the original domain, an assumption fulfilled in the case of finite groups (the root ring represents a finite extension). For the existence of the rational basis I give a proof valid for arbitrary fields as coefficient domain; hence I can prove the finiteness criterion under the above restriction.

That, for invariants of finite groups, the finiteness criterion is fulfilled rests on the principle of symmetric functions. Whereas in characteristic zero the coefficients of the Galois resolvent already furnish the integral basis, in the general case they generate only the finite subring required in the finiteness criterion. From the existence of this finite subring the finiteness of the “relative” invariants and of the “modular” invariants follows by the same arguments.

The invariants considered here include, as special cases, the “projective invariants mod p ” treated by Dickson and his school.⁵ Here the finiteness theorem was known for modular invariants, but not for general “formal” invariants.

§ 1. The Finiteness Criterion

1. Theorem on the existence of the rational basis. First formulation: Every system S of rational functions of n indeterminates, with coefficients from a field P , possesses a finite rational basis; that is, there are finitely many functions of the system such that every function of the system can be expressed rationally, with coefficients from P , through these finitely many.

Second formulation: Let $\mathfrak{K}$ be an intermediate field between P and $P(x_1, \dots, x_n)$ – where $x_1, \dots, x_n$ denote indeterminates – of transcendence degree $t \leq n$; let $y_1, \dots, y_t$ be an irreducible system⁶ from $\mathfrak{K}$. Then $\mathfrak{K}$ becomes a finite algebraic extension of $P(y_1, \dots, y_t)$.

The two formulations are in fact equivalent. It is enough to prove the existence of the rational basis for all intermediate fields; for the field derived from a system S becomes such an intermediate field $\mathfrak{K}$, whose rational basis immediately gives a rational basis of S , since every element of $\mathfrak{K}$ becomes a rational combination of finitely many functions from S . From the second formulation it follows, however, that an arbitrary irreducible

²For an elementary proof of finiteness see E. Noether, Der Endlichkeitssatz der Invarianten endlicher Gruppen. Math. Ann. 77 (1916), pp. 89–92.

³For the concepts of field theory see E. Steinitz, Algebraische Theorie der Körper. J. f. M. 137 (1910), pp. 167–309. Characteristic in § 4, root field in § 12, quotient field in § 3.

⁴Compare E. Noether, Algebraische und Differentialinvarianten. Jahresber. d. d. Math.-Ver. 32 (1923), pp. 177–184, no. 3, Theorem 4, and the literature cited there. In the last sentence of the criterion there it incorrectly says “module basis with respect to $\mathfrak{R}$ ” (fundamental system of integral quantities) instead of “with respect to a certain subring of $\mathfrak{R}$ ”.

⁵Compare Dickson, On Invariants and the theory of Numbers. The Madison Colloquium 1913. More recent literature is in the volumes of the Transactions of the American Mathematical Society that have appeared since then.

⁶A system is called irreducible, according to Steinitz, if it consists of algebraically independent functions. A system is said to have transcendence degree t if it contains at least one irreducible system of t elements, but none of more elements. For the simple theorems on irreducible systems used below, see Steinitz, § 22.

system $y_1, \dots, y_t$ from $\mathfrak{K}$, enlarged by the finitely many functions that characterize $\mathfrak{K}$ as a finite extension of $P(y_1, \dots, y_t)$, gives such a rational basis. Conversely, if according to the first formulation a rational basis of $\mathfrak{K}$ is given, then by the definition of transcendence degree each basis element must be algebraically dependent on $P(y_1, \dots, y_t)$; hence $\mathfrak{K}$ is thereby characterized as a finite algebraic extension of $P(y_1, \dots, y_t)$.

The theorem will be proved in the second formulation. First let $\mathfrak{K}$ have the same transcendence degree n as $P(x_1, \dots, x_n)$; let $y_1, \dots, y_n$ be an irreducible system from $\mathfrak{K}$ and consequently from $P(x_1, \dots, x_n)$. Because the transcendence degree of $y_1, \dots, y_n$ is n , each x_i becomes algebraic over $P(y_1, \dots, y_n)$; thus $P(x_1, \dots, x_n)$, and hence $\mathfrak{K}$ as an intermediate field, becomes a finite algebraic extension of $P(y_1, \dots, y_n)$.

Now let $t < n$; let the numbering of the x be chosen so that $y_1, \dots, y_t, x_{t+1}, \dots, x_n$ form an irreducible system from $P(x_1, \dots, x_n)$. Then $x_{t+1}, \dots, x_n$ are irreducible with respect to $P(y_1, \dots, y_t)$ and hence, by the transitive law of algebraic dependence, also irreducible with respect to $\mathfrak{K}$ and to every intermediate field $\mathfrak{M}$ between $P(y_1, \dots, y_t)$ and $\mathfrak{K}$. This means that $\bar{\mathfrak{K}} = \mathfrak{K}(x_{t+1}, \dots, x_n)$, respectively $\bar{\mathfrak{M}} = \mathfrak{M}(x_{t+1}, \dots, x_n)$, becomes a purely transcendental extension of $\mathfrak{K}$, respectively $\mathfrak{M}$; hence every element of $\bar{\mathfrak{K}}$ not belonging to $\mathfrak{K}$ is transcendental with respect to $\mathfrak{K}$, and similarly every element of $\bar{\mathfrak{M}}$ not belonging to $\mathfrak{M}$ is transcendental with respect to $\mathfrak{M}$. Likewise let $\bar{P}$ be set equal to the purely transcendental extension $P(x_{t+1}, \dots, x_n)$; then $y_1, \dots, y_t$ are irreducible with respect to $\bar{P}$.

The proof can now be reduced to the case already settled by taking $\bar{P}$ as coefficient domain. From $P < \mathfrak{K} < P(x_1, \dots, x_t, x_{t+1}, \dots, x_n)$ ⁷ it follows that $\bar{P} < \bar{\mathfrak{K}} < \bar{P}(x_1, \dots, x_t)$, where $\bar{\mathfrak{K}}$ has the same transcendence degree t over $\bar{P}$ as $\bar{P}(x_1, \dots, x_t)$. Hence $\bar{\mathfrak{K}}$ has a finite rational basis with respect to $\bar{P}$ as coefficient domain, say $y_1, \dots, y_t; z_1, \dots, z_s$. Here the z can be chosen from $\mathfrak{K}$, since $\bar{\mathfrak{K}}$ is the union field of $\mathfrak{K}$ and $\bar{P}$. It remains to show that $\bar{\mathfrak{K}}$ equals $\bar{\mathfrak{M}} = P(y_1, \dots, y_t; z_1, \dots, z_s)$. Now $\bar{\mathfrak{M}}$ is an intermediate field between $P(y_1, \dots, y_t)$ and $\bar{\mathfrak{K}}$; hence every element of $\bar{\mathfrak{K}}$ is algebraic with respect to $\bar{\mathfrak{M}}$. On the other hand, $\bar{\mathfrak{M}}$ equals $\bar{\mathfrak{K}}$, and hence $\bar{\mathfrak{K}}$ is contained in $\bar{\mathfrak{M}}$. Thus $\bar{\mathfrak{K}}$ is an intermediate field between $\bar{\mathfrak{M}}$ and $\bar{\mathfrak{M}}$; since every element of $\bar{\mathfrak{M}}$ not belonging to $\bar{\mathfrak{M}}$ is transcendental with respect to $\bar{\mathfrak{M}}$, it necessarily follows that $\bar{\mathfrak{K}}$ is identical with $\bar{\mathfrak{M}}$.

Corollary: If $P < \mathfrak{K} < \mathfrak{L} < P(x_1, \dots, x_n)$ and $\mathfrak{L}$ is algebraic with respect to $\mathfrak{K}$, then $\mathfrak{L}$ is a finite algebraic extension of $\mathfrak{K}$. For every element of a rational basis of $\mathfrak{L}$ is then algebraic with respect to $\mathfrak{K}$; hence $\mathfrak{L}$ is characterized as a finite extension of $\mathfrak{K}$.

2. Proof of the finiteness criterion. The condition is plainly necessary; for if $\mathfrak{S}$ is a finite integral domain, then $\mathfrak{S}$ itself can be regarded as the finite subring $\mathfrak{R}$ occurring in the criterion.

The condition is sufficient under the assumption that the root field $P^{1/p}$ is a finite extension of the coefficient field⁸ whenever the latter has characteristic p ; for characteristic zero no restrictive assumption is needed. To prove this one has to show that $\mathfrak{S}$ has a finite module basis with respect to a subring of $\mathfrak{R}$.

Let $\bar{\mathfrak{K}}$ denote the quotient field⁹ of $\mathfrak{R}$, and let $\bar{\mathfrak{L}}$ denote the quotient field of $\mathfrak{S}$. These quotient fields exist since the rings involved have no zero divisors; and one has $P < \bar{\mathfrak{K}} < \bar{\mathfrak{L}} < P(x_1, \dots, x_n)$. Hence by the corollary under 1., $\bar{\mathfrak{L}}$ becomes a finite algebraic extension field of $\bar{\mathfrak{K}}$. Moreover, by hypothesis every element of $\mathfrak{S}$ is integral with respect to $\mathfrak{R}$, and thus $\bar{\mathfrak{S}}$ becomes a subring of the ring $\bar{\mathfrak{S}}'$ consisting of all elements of $\bar{\mathfrak{L}}$ integral over $\bar{\mathfrak{K}}$. But since nothing is assumed about the integral closedness of $\mathfrak{R}$ in $\bar{\mathfrak{K}}$, the fact that the divisor-chain theorem is preserved under finite extension cannot be applied directly. Rather, one must first pass to a likewise finite subring $\mathfrak{T}$ of $\mathfrak{R}$ for which this integral closedness is fulfilled.

First suppose that the coefficient domain P contains infinitely many elements. Then one can select from $\mathfrak{R}$ an irreducible system $g_1(x), \dots, g_t(x)$ in such a way that $\mathfrak{R}$, and hence $\mathfrak{S}$, is integral over $\mathfrak{T} = P[g_1(x), \dots, g_t(x)]$. Namely, let $f_1(x), \dots, f_k(x)$ be the integral basis of $\mathfrak{R}$ existing by hypothesis; if this does not form an irreducible system, then – after a possible renumbering of the indices – let $F(f_k; f_1, \dots, f_{k-1}) = 0$ be a relation satisfied by f_k . If one replaces $f_1, \dots, f_{k-1}$ by

$$f_1 + t_1 f_k, \dots, f_{k-1} + t_{k-1} f_k,$$

then f_k , and hence also $f_1, \dots, f_{k-1}$, is integral over these new functions when the t_i are chosen as indeterminates. Since P has infinitely many elements, the t_i can be specialized to elements of P in such a way that this integral dependence is preserved. After finitely many steps, using the transitive law of integral dependence, one therefore arrives at the required ring $\mathfrak{T} = P[g_1(x), \dots, g_t(x)]$. Then $\bar{\mathfrak{L}}$ is a finite extension field of the quotient field $P(g_1(x), \dots, g_t(x))$ of $\mathfrak{T}$, and the ring $\bar{\mathfrak{S}}'$ of all elements of $\bar{\mathfrak{L}}$ integral over $\bar{\mathfrak{K}}$ is identical with the ring of all elements of $\bar{\mathfrak{L}}$ integral over $\mathfrak{T}$.

⁷ $P < \bar{\mathfrak{K}}$ means: P is contained in $\bar{\mathfrak{K}}$, hence is a subfield of $\bar{\mathfrak{K}}$.

⁸ See p. 28, note 2.

⁹ See p. 28, note 2.

In $\mathfrak{T}$ the divisor-chain theorem holds for the system of all ideals, and $\mathfrak{T}$ is integrally closed in its quotient field, both because $\mathfrak{T}$ is isomorphic to the polynomial domain in t indeterminates. From the assumption that $P^{1/p}$ is finite over P it follows further that $\mathfrak{T}^{1/p}$ is finite over $\mathfrak{T}$.¹⁰¹¹ Hence in $\mathfrak{S}'$ the divisor-chain theorem holds for the system of all $\mathfrak{T}$ -modules; in particular, $\mathfrak{S}$, as a subring of $\mathfrak{S}'$ containing $\mathfrak{T}$, has a finite $\mathfrak{T}$ -module basis. The existence of the finite $\mathfrak{T}$ -module basis for $\mathfrak{S}$ continues to hold without any restrictive assumption on the coefficient domain P if $\mathfrak{L}$ is an extension of the first kind of $P(g_1(x), \dots, g_t(x))$, hence in particular always when P has characteristic zero.¹²

Let $h_1(x), \dots, h_s(x)$ be a $\mathfrak{T}$ -module basis of $\mathfrak{S}$. Then the representation

$$f(x) = A_1(g)h_1(x) + \dots + A_s(g)h_s(x)$$

for an arbitrary $f(x)$ from $\mathfrak{S}$ – where the A , as elements of $\mathfrak{T}$, denote polynomials in the g – shows that $g_1(x), \dots, g_t(x), h_1(x), \dots, h_s(x)$ form the required integral basis.

If the field P contains only finitely many elements, then the field $P(u)$ obtained by adjoining an indeterminate u – that is, an element transcendental over $\mathfrak{S}$ – contains infinitely many elements. Thus the ring $\mathfrak{S}(u)$ has a finite integral basis with respect to $P(u)$ as coefficient domain; and since $\mathfrak{S}(u)$ is the union ring of $\mathfrak{S}$ and $P(u)$, this basis can be chosen from $\mathfrak{S}$ by splitting according to powers of u , although the basis elements g_i of $\mathfrak{S}$ will now no longer form an irreducible system. Accordingly every polynomial $f(x)$ from $\mathfrak{S}$ admits a representation $C(u)f(x) = G(u, g_i, h_i)$, where $C(u)$ denotes a nonzero polynomial from $P[u]$ and G becomes integral in u, g_i, h_i . Comparing the coefficients of a suitable power of u shows that $g_i(x)$ and $h_i(x)$ give an integral basis of $\mathfrak{S}$. Hence the **finiteness criterion is proved in full generality**.

§ 2. Finiteness of the Invariants

1. By a **finite linear group of characteristic p** is meant a group $\mathfrak{H}$ of h linear substitutions $A_1, \dots, A_h$ (of nonzero determinant) with coefficients from a field P of characteristic p . Here A_k is the linear substitution

$$x_1^{(k)} = a_{11}^{(k)}x_1 + \dots + a_{1n}^{(k)}x_n, \dots, x_n^{(k)} = a_{n1}^{(k)}x_1 + \dots + a_{nn}^{(k)}x_n,$$

or, in abbreviated form, $(x^{(k)}) = A_k(x)$. Thus the group $\mathfrak{H}$ carries the row (x) with elements $x_1, \dots, x_n$ into the rows $(x^{(k)})$ with elements $x_1^{(k)}, \dots, x_n^{(k)}$. Since the identity must be contained among $A_1, \dots, A_h$, the row (x) is also contained among the rows $(x^{(k)})$.

By an **integral rational (absolute) invariant of the group** is meant such an integral rational function of $x_1, \dots, x_n$ – with coefficients from P^{13} – that remains identically unchanged under application of $A_1, \dots, A_h$. Thus all symmetric functions, with coefficients from P , of the rows $(x^{(k)})$ belong to these invariants. Conversely, for every such invariant one has

$$f(x) = f(x^{(1)}) = \dots = f(x^{(h)}) \quad \text{or} \quad h \cdot f(x) = f(x^{(1)}) + \dots + f(x^{(h)}).$$

It follows that in the case where the order h of the group is not divisible by the characteristic – in particular in characteristic zero – the symmetric functions of the rows $(x^{(k)})$, with coefficients from P , exhaust the totality of invariants. Since these symmetric functions possess a finite integral basis, the proof of finiteness is accomplished in this case; at the same time the integral basis is obtained as the system $G_{\alpha\alpha_1\dots\alpha_n}(x)$ of coefficients of the Galois resolvent:¹⁴

$$\begin{aligned} \Phi(z, u) &= \prod_{k=1}^h (z - u_1x_1^{(k)} - \dots - u_nx_n^{(k)}) \\ &= z^h + \sum G_{\alpha\alpha_1\dots\alpha_n}(x) z^\alpha u_1^{\alpha_1} \dots u_n^{\alpha_n} \quad (\alpha \neq h; \alpha + \alpha_1 + \dots + \alpha_n = h). \end{aligned}$$

2. In the case where h is divisible by p , it is not necessary that every invariant of the group be a symmetric function of the rows $(x^{(k)})$, since one can no longer conclude from $h \cdot f(x)$ to $f(x)$. But the ring $\mathfrak{R}$ derived

¹⁰Compare Hilbert, Über die vollen Invariantensysteme, § 1. Math. Ann. 42 (1893), pp. 313–373.

¹¹Compare the note of Artin and van der Waerden cited in the introduction.

¹²Compare, for example, E. Noether, Abstrakter Aufbau der Idealtheorie in algebraischen Zahl- und Funktionenkörpern, § 3. Math. Ann. 96 (1926), pp. 26–61.

¹³This is not a restriction of generality inasmuch as every invariant with coefficients from a field of characteristic p – which must possess a common overfield with P in order for the invariants to be defined – can be linearly decomposed into finitely many invariants with coefficients from P . In the general case one need only express the coefficients through ones that are linearly independent over P ; then the property of invariance is satisfied identically in these independent coefficients.

¹⁴See p. 28, note 1.

from the finitely many invariants $G_{\alpha\alpha_1\dots\alpha_n}(x)$ – the coefficients of the Galois resolvent – forms a finite subring of the system $\mathfrak{S}$ of all invariants, and it remains to show that $\mathfrak{S}$, with respect to $\mathfrak{R}$, satisfies the hypotheses of the finiteness criterion.

First, without loss of generality,¹⁵ the subfield $\overline{P}$ of P derived from all substitution coefficients $a_{\mu\nu}^{(k)}$ may be taken as coefficient domain. This field, derived from finitely many elements, is thus obtained by adjoining finitely many algebraic or transcendental elements to the prime field; and since the prime field is perfect, it follows that $\overline{P}^{1/p}$ is finite with respect to $\overline{P}$.¹⁶

It remains to show that $\mathfrak{S}$ depends algebraically and integrally on $\mathfrak{R}$. By 1., among the h rows $x^{(k)}$ there also occurs the original $x_1, \dots, x_n$; hence the Galois resolvent $\Phi(z, u)$ vanishes identically in u for $z = u_1x_1 + \dots + u_nx_n$. If one specializes in each case all u to zero except u_i , which is set equal to one, one obtains a relation

$$x_i^h + \sum x_i^\alpha G_{\alpha 0 \dots \alpha_i \dots 0}(x) = 0 \quad (\alpha \neq h; \alpha + \alpha_i = h),$$

which shows that each x_i depends algebraically and integrally on $\mathfrak{R}$. But then the same holds for every polynomial in x ; in particular $\mathfrak{S}$ is integral over $\mathfrak{R}$. Thus the hypotheses of the finiteness criterion are fulfilled, and the **finiteness of the invariants is proved**.

3. It should be noted that the same arguments give the **finiteness of the relative and modular invariants**. A polynomial $f(x)$ is called a **relative invariant** of the group if all $f(x^{(k)})$ differ from $f(x)$ only by a nonzero factor from P . The system of absolute invariants – in particular the ring $\mathfrak{R}$ derived from the $G_{\alpha\alpha_1\dots\alpha_n}(x)$ – therefore forms a finite subring of the ring derived from all relative invariants; and, as above, the hypotheses of the finiteness criterion with respect to $\mathfrak{R}$ are fulfilled.

A polynomial $f(x)$ is called a **modular invariant** of the group if $f(x)$ becomes equal to $f(x^{(k)})$ for every special value system $x_1 = \alpha_1, \dots, x_n = \alpha_n$ from P . Every absolute invariant is at the same time a modular invariant; if, however, P contains only finitely many elements, the converse need not hold. Here too the hypotheses of the finiteness criterion are fulfilled with respect to the subring $\mathfrak{R}$ derived from the $G_{\alpha\alpha_1\dots\alpha_n}(x)$.

¹⁵See p. 33, note.

¹⁶See p. 32, note 2.